ReNu-Biome portal — requirements: Phase 1 fixes and Phase 2

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Purpose and how to read this

This is the single requirements document for the ReNu-Biome customer portal, written so the developer can estimate and build from it. Part A is the Phase 1 fix list (the February 2027 release) rolled in from the October 9 review and the owner's answers. Part B is Phase 2: transferring hosting and ownership to ReNu-Biome, OptimoRoute delivery notifications, Otodata tank fill monitoring, nutrition programs per ranch, soil and tissue sample submission for custom recommendations, installing the app from a QR code, taking the app from demo to installed for the CEO and the first customer, the sign-in and Home screens built to the approved prototype, and what the owner's program spreadsheet means for the app. Before either part: the Phase 1 items still open from the October 19 check of the build. Suggested order: Part A, then hosting, then install and pilot, then the four Phase 2 features. Prototype v2 is the visual reference for every screen in both parts.

It builds on the Phase 1 fix list (data model, deliveries, invoices, admin console) and assumes that work is done or in progress. Where a feature here depends on a Phase 1 item, it says so. Each feature states what the user should be able to do, how it connects to the outside system, and what "done" looks like. The acceptance tests at the end are the definition of done for the fixed price.

Current build, checked October 19

The developer has rebuilt the app on the Phase 1 list since the October 9 review, and the new features below can be specified against what now exists rather than against the old prototype.

Area	What is there now	Matters for
Sign-in	Email and password only; Sign Up, Apple and Google buttons removed; Employee role removed; demo buttons labelled "staging only"	Hosting transfer must keep demo logins off production

Area	What is there now	Matters for
Data model	Ranch → blocks (block ID, crop, variety, rootstock, acreage, planting year, irrigation type, fertigation set, county, APNs) and ranch → tanks (name, capacity, location, level, pressure); acreage computed; no map; no entity layer	Programs attach to ranches and blocks; Otodata attaches to tanks
Deliveries	Renamed from Orders; fields for product, gallons, ranch, tank, blocks, delivery window, gate code, contact on arrival, notes; statuses Scheduled → Out for delivery → Delivered, plus Cancelled; three-stage timeline	OptimoRoute drives these statuses
Invoices	Number, dates, amount, status Open / Paid, paid date, delivery link, product, gallons, blocks, an empty quickbooksInvoiceId field; no PDF attached in the demo data	Unchanged by this phase
Home	Driven by live data: acres, ranches and blocks, active deliveries, open invoices, delivery schedule, ranches	Tank levels and program stage can be added here
Admin console	Today's Operations with deliveries due today and tomorrow, overdue invoices, Create Order; tabs for Deliveries, Customers, Ranches, Invoices, Products, Settings	Admin screens for programs and sample review go here
Feature flags	/api/app/feature-flags with global and per-customer values for payments, recommendations, labResults, documents, compliance, messaging, staffTasks, optimoroute, telemetry, quickbooks, map, all off	OptimoRoute and Otodata switch on the existing optimoroute and telemetry flags
Offline	Web app manifest and service worker present; ranches, deliveries and invoices cached in local storage	Unchanged

Area	What is there now	Matters for
Lab results	/api/app/lab-results exists and returns an empty list; no screen	Starting point for sample submission

Not yet seen in this check: invoice PDF upload (demo invoices carry no PDF), the products catalogue (the demo list holds one test product named "SRavanga"), and the Phase 1 acceptance script has not been run. Those are Phase 1 closure items, not part of this document.

Phase 1 items to close first

Checked against the Phase 1 list on October 19. Most of Phase 1 is done; these remain and come before any Phase 2 work. Items 1 to 3 are small and item 1 hides most of the admin work that already exists.

#	Finding	What to do	Phase 1 item
1	The admin Customers tab is empty and every delivery card reads "Customer" and "Ranch — Tank" instead of names. Cause: the app calls /api/admin/customers (404) while the backend serves /api/app/admin/customers (200).	Fix the path; show customer, ranch and tank names on Today and Deliveries; confirm Add Customer, Add Ranch and the login creation work from the console	Customers, Customer login
2	No invoice PDF. The Mark Delivered form takes number, amount and dates but has no file field; demo invoices carry no PDF and the customer's View Invoice has nothing to open	Add PDF upload to invoice entry and to the admin Invoices tab; store the file with the invoice; open it in the app	Invoice entry, Customer invoice list
3	Every tab's content stays in the page; the More screen contains Home, Ranches, Deliveries and Invoices too	Render only the active screen or use a navigator that unmounts inactive tabs	Unmount screens

#	Finding	What to do	Phase 1 item
4	Email sending not evident: no email provider in the build and no sending account exists, so delivery status emails may not be wired	Set up the email account (section 1.6) and prove a status email arrives in a real inbox	Email updates, Email sending
5	Product catalogue is the mock list (Biome Care, N-CARE, K-RUSH, Bee Bloom, plus a test entry "SRavanga")	Replace with the real products from the owner's program spreadsheet (section 9), with analysis and density fields	Products
6	No security headers; no rate limit on login (eight bad attempts accepted); token life 7 days	Add headers, rate-limit /api/auth/login, shorten the token with a refresh token. The MFA email code is already built, which is ahead of plan	Hardening (moved to Phase 2 in the sent list; do it with section 1)
7	Not verifiable from outside: whether data survives a deploy, backups, monitoring, staging vs production, the README	Covered by section 1; the developer demonstrates each at hand-over	Hosting, handover
8	Ionicons font 404s on load; Create Order opens the Deliveries tab rather than the form	Bundle the font; open the form directly	Console errors, Create order

Once 1 to 5 are done, the office runs the Phase 1 acceptance script with the developer on a call, on real data, before section 7 (pilot) starts.

Part A. Phase 1 fixes (February 2027 release)

Rolled in from the October 9 fix list so the developer has one document. These are the items for the February 2027 release before bloom: the data model, deliveries, invoices, the admin console, quality fixes, the feature-flag system and hosting. Requirement codes (FLD-1, ORD-3 and so on) refer to Customer_Portal_Requirements_v1.2.docx. Sections

here are numbered A1 to A11 so they do not collide with the Phase 2 sections in Part B. Build the data model first; most later items depend on it. Prototype v2 (section 8 in Part B) is the visual reference for every screen in this part too.

Owner decisions that set the Phase 1 scope

Topic	Decision	Effect on the build
Customers	One legal entity and one user per customer; every account created by ReNu-Biome	No entity switcher, no customer-managed users, no sign-up, no split billing
Language	English only; the owner is the user	No Spanish
Products	The website's product lines are the catalogue; sold in gallons; no prices shown anywhere in the app	Catalogue without prices; no pricing admin
Ordering	Customers do not order through the app at launch; ReNu-Biome enters orders from phone calls	Order request form, shop, cart, reorder and cancel go behind a feature flag
Delivery	Into the customer's tank; routed with OptimoRoute	Delivery schedule and status in the app; OptimoRoute integration in Part B
Status updates	Each customer may choose channels later; email only at launch	Email in Phase 1, with per-customer text and push options in Phase 2
Accounting	QuickBooks Online, kept by the CEO; no connection at launch	Invoices entered by hand; QBO sync behind a flag
Payment	On account after delivery, by check; no payments provider yet	In-app ACH and card behind a flag
Statements	Year-end purchase summary	Built; no monthly statements or reminders

Topic	Decision	Effect on the build
Agronomy, lab results, documents, e-signature	Wanted later	Behind feature flags; samples and recommendations are Part B section 5
Staff	Owner plus one office person; no employee tasks for now	Two admin logins; task module behind a flag
Admin first screen	Deliveries due today	Today view
Contact	Customers email the office at support@renu-biome.com; no phone number or hours in the app	Email button; in-app thread behind a flag
Weather	At the ranch	Weather by ranch location
Map	None	The photo map is removed, not replaced
Launch	February 2027, before bloom; fixed-price developer; ReNu-Biome owns the app afterwards	Scope below; handover required

Still open: which monthly reports the owner wants (section A6, item A6.7).

Phase 1 scope

What a customer needs on day one: their ranches and blocks, their delivery schedule with email updates, their invoices, and an admin console the office can run. The sign-in and Home screens are built to prototype v2 (Part B, section 8); everything the prototype marks Phase 2 is left out of this release.

A1. Data model (FLD-1)

Everything below depends on this. Today the model is Customer → Ranch → Tank. It must become Customer → Ranch → Block, with tanks attached to ranches. There is no entity layer: one customer is one billing account.

A1.1 Add blocks under ranches. A block is a planted unit the customer tracks deliveries against. Fields: block ID (the customer's own, for example N-4), crop (almond, pistachio, grape), variety and rootstock, acreage, planting year, irrigation type, fertigation set or valve, county, APN(s) (several per block). Done: the Ranch screen lists its blocks in a table (block, crop and variety, acres, planted); tapping opens a block detail with those fields and a timeline of deliveries.

A1.2 Tanks stay, attached to ranches. Keep the tank register (name, capacity, location) on the ranch screen below the blocks. Tank level and pressure fields can stay but are optional until telemetry exists.

A1.3 Acreage is computed. The ranch card says 85 acres, the entity sheet 350 and the total 1,250. Ranch acreage is the sum of its blocks; never typed separately.

A1.4 Remove the map. Delete the "Interactive Map" screen, the stock wheat-field photo, the pin-dropping and the Add Tank pin flow. Tanks are added and edited from a plain form on the ranch screen. No map of any kind is in the launch scope; if one is wanted later it becomes a flag.

A1.5 Remove the entity switcher. The "Select Operating Entity" sheet and the "All entities" button go. The Home header shows the customer's name.

A2. Deliveries (ORD-3, ORD-4)

Customers phone in orders; ReNu-Biome enters them. The customer's side is a delivery schedule and status, not a store. Rename the Orders tab to Deliveries.

A2.1 Admin creates the order. From the admin console, the owner or office person creates an order for a customer: product (from the catalogue), quantity in gallons, ranch and tank, blocks it is for (multi-select), requested delivery window (date range plus morning or afternoon), gate code and who to call on arrival, notes. Customer pricing is not entered or shown; the amount appears only on the invoice the office enters later.

A2.2 Statuses. Scheduled (date set) → Out for delivery → Delivered, plus Cancelled with reason, set by admin only. Remove the current five-step timeline that assumes payment before dispatch (Order Placed, Accepted and Payment Sent, Payment Confirmed, Dispatched, Delivered); keep the timeline component with these stages. Customers cannot cancel.

A2.3 Customer view. The Deliveries tab lists upcoming deliveries first (date, program name, gallons, tank, blocks, status) then past ones, filterable by ranch, as in prototype v2. Tapping one shows the detail with the product and its analysis and, once delivered, the

delivered quantity and date and a link to its invoice.

A2.4 Email updates. On Scheduled, Out for delivery and Delivered, send the customer an email (set up under A10.1b) with the product, gallons, tank and date. Email is the only customer channel in Phase 1; text and push choices come in Phase 2 (section A7). Admin gets an email when a delivery is marked Delivered by the other admin.

A2.5 Remove customer ordering from the launch build. Hide the Shop tab, the cart, "Request an order", "Reorder" and "Cancel" behind the ordering feature flag (section A9). The code can stay; the customer must not see it.

A2.6 OptimoRoute (later). ReNu-Biome routes deliveries with OptimoRoute. Design the order record so an order can carry an OptimoRoute order ID and receive status and delivery time from it later; the integration itself is Part B section 2.

A3. Invoices (FIN-1, FIN-7)

Invoices today are mock records with a PDF link that does not resolve. At launch the office enters each invoice by hand from QuickBooks; the QuickBooks connection itself is deferred (flag, section A9).

A3.1 Invoice entry by admin. When a delivery is marked Delivered, the admin console asks for the invoice number, invoice date, due date, amount and the invoice PDF exported from QuickBooks. Admin can also add an invoice not tied to a delivery, and edit or void one. No prices or line pricing are computed by the app.

A3.2 Paid status by admin. When the CEO records a check in QuickBooks, the office marks the invoice Paid in the app with the payment date. Statuses: Open, Overdue (past due date), Paid, Void.

A3.3 Customer invoice list. Filter Open, Paid, All. Each invoice shows number, date, due date, amount, status, and the delivery it belongs to with product, gallons and blocks. The PDF opens in the app from the stored file; remove the hardcoded renubiome.com PDF URL.

A3.4 Year-end purchase summary. For a chosen year, a PDF and CSV per customer listing every delivery and invoice with product, gallons and amount, totalled, for the customer's accountant. Available to the customer under More and to admin per customer.

A3.5 Ready for QuickBooks later. Store a QuickBooks invoice ID and customer ID field on invoices and customers now, left empty, so the sync can be added without migration.

A6. Admin console (ADM-1–4)

Two people use it: the owner and an office person, from a phone or laptop. The owner must be able to maintain customers, products and documents without calling the developer. Prototype v2's admin console (Part B, section 8.7) is the reference.

A6.1 Today view. The first screen is deliveries due today and tomorrow (customer, program name, gallons, ranch, tank, window, gate code, who to call, as bullet points) with a one-tap Mark delivered that asks for the delivered gallons and then the invoice details (A3.1). Above it, metric tiles with icons: due today and tomorrow, open this week, overdue invoices with the dollar total. Replace the static "$42,500 revenue, +14%" tiles with values computed from the app's own records.

A6.2 Customers. The list is empty today although three mock customers exist. Show every customer with contact, ranches, acres, year-to-date gallons and open balance. From a customer: create or edit ranches, blocks and tanks one at a time; create an order; reset the password. Spreadsheet import, view as customer and invitations are Phase 2.

A6.2a Customer login, created by admin. There is no self-service sign-up in this phase. Admin creates the customer's single login from the customer screen: name, email, mobile number and a password the admin types; admin gives the password to the customer by phone. The customer can change it under More. Admin can reset it. The Sign Up link and the Apple and Google buttons are removed from the sign-in page now, even though the rest of accounts and security is Phase 2, because otherwise they remain the only way in.

A6.3 Create order. The form in A2.1, reachable from Today and from a customer.

A6.4 Products. Edit name, description, analysis (percent N, P_2O_5, K_2O, Ca and lb per gallon, see Part B section 9.1), OMRI flag and image. No price field anywhere in the app. Product documents (labels, SDS, certificates) are behind the documents flag.

A6.5 Deliveries. All orders with filters by status, customer and date; edit, reschedule, cancel with reason, mark delivered.

A6.6 Invoices. Invoice list across customers with status; add, edit, void, mark paid; upload the PDF; link an invoice to a delivery.

A6.7 Reports. Decided: four monthly reports, each on screen for a chosen month or year to date and downloadable as PDF and CSV: sales by customer (invoice totals), sales by product (gallons and invoice totals), receivables aging (current, 30, 60, 90+ days), and gallons by month (by product and crop). Plus CSV export of deliveries and invoices by date range. Sales figures come from the invoice totals the office enters; the app stores no prices. Prototype v2: admin More → Reports.

A6.8 Notifications to admin. An email to the owner and office person when a delivery is marked delivered by the other admin. Text and push for admin come with Phase 2.

A8. UX and quality (UX-1–3, UX-8)

A8.1 Installable and offline. Add a web app manifest and service worker so the app installs to the home screen with the ReNu-Biome icon and supports push. Cache the last-loaded Home, Ranches, blocks, Deliveries and Invoices for reading without signal.

A8.2 Unmount screens you leave. Every tab's content stays in the DOM (the page text for More contains Home, Ranches, Orders, Invoices, Messages and Shop). Render only the active screen or use a navigator that unmounts inactive tabs.

A8.3 Loading states. Lists take 5–8 seconds on first load and show a bare spinner. Show skeleton placeholders as in the prototype; cache the last response so a second visit is instant.

A8.4 Truncated labels. At phone width buttons read "Live…", "R.", "C.", "Reor…", "Interactiv…". Use labels that fit 375 px.

A8.5 Bundled assets. Icon fonts load from cdnjs and photos from images.unsplash.com; both fail without signal and the Unsplash images need licence review. Bundle the icon font; use ReNu-Biome's own product and ranch images.

A8.6 Test data. Order ORD-1097 shows cancellation reason "fghdhsrhh". Seed data must read like real records or not exist.

A8.7 Console errors. Six 404s on load and an Animated useNativeDriver warning. Clear them.

A8.8 Field use. Minimum 48 px tap targets, readable in bright sun, dynamic text size.

A8.9 Print and PDF. Invoices and the year-end summary print cleanly and export to PDF.

A8.10 No Spanish. Remove any partial localisation work; English only.

A9. Build behind a feature flag

Each item below is wanted later. Build the data model so it can be added without migration, keep any existing code, and hide it behind a named flag the owner can switch per customer or globally from the admin console. Nothing in this section is in the February release and none of it is visible to customers until switched on. Part B turns on four of them: recommendations, labResults, optimoroute and telemetry.

Flag	What it turns on	What exists today	Notes
ordering	Customer order requests, Shop, cart, reorder; admin confirms requests	Shop, cart, order status, reorder, cancel	Keep the request-then-confirm flow; still no prices shown
payments	Pay invoices in-app by ACH and card through a payments provider; status back to QuickBooks	Emailed payment link	Provider account in ReNu-Biome's name; card and bank numbers never stored by the app
recommendations	A recommendation written by ReNu-Biome from a sample, shown on the block and saved as a program change	Nothing (API returns 404)	Part B section 5
labResults	Customer sends soil, tissue and water results per block; values against UC reference ranges; trend per nutrient	A per-ranch soil form in the Employee app saved by /api/app/lab-results. Not per block, no file upload, no reference ranges	Part B section 5: key by block, accept PDF or photo, sampleType, store the real nutrients with sample date
documents	Product labels, SDS, OMRI and CDFA certificates attached to deliveries; contracts sent for e-signature; customer uploads; audit pack export	Nothing	The owner wants e-signature; certificates and labels to be supplied by the owner

Flag	What it turns on	What exists today	Notes
compliance	Applied nitrogen per APN and season for ILRP coalition reports	Nothing	Needs blocks with APN and yield, which the data model carries; defined in Part B section 9.4
messaging	In-app message thread per customer with photo from the field; SMS and email replies land in the thread	One-way inbox and a compose button	Owner answers; no expected response time shown
staffTasks	Employee role, assigned tasks (tank maintenance, sampling visits)	A full Field Staff app with static tiles and a task list; /api/app/applications is 404	Scope creep for now. Hide the role at login. The employee token today can read every customer's orders and invoices; when the role returns, restrict it to tasks, ranches and lab entry
optimoroute	Delivery status and ETA from OptimoRoute; proof of delivery with photo, quantity, time and location	Nothing	Part B section 2; store an OptimoRoute order ID on each order now
telemetry	Tank levels and alerts from Otodata monitors	Mock level and pressure fields	Part B section 3

Flag	What it turns on	What exists today	Notes
quickbooks	QuickBooks Online sync of customers, products, invoices and payments; sync-error screen	Nothing	Fields for QuickBooks IDs exist from launch (A3.5); the CEO authorises the connection when it is built
map	Ranch map on satellite tiles with block boundaries and tank markers	A stock photo with draggable pins, removed at launch	Not wanted now; blocks carry county and APN so boundaries can be imported later

The admin console needs a Feature flags screen listing these with on/off per customer and a global default (prototype v2: admin More → Feature flags).

Employee app, reviewed October 9. The Emp demo login opens a separate Field Staff app, not just a role. Keep the task and lab-results code behind the staffTasks and labResults flags. The employee role must not be able to read customer invoices and orders, which it can today. During the review a demo soil record was submitted by accident from the employee account (pH 6.8, N 42 ppm, dated October 10); delete it with the other seed data.

A10. Hosting, handover and what ReNu-Biome provides

Part B section 1 (transfer of hosting and ownership) supersedes this section where they overlap; the items here are the minimum for the February release.

A10.1 Hosting, set up by the developer. Both services run on Render's free tier with 5–8 second cold starts. Before the February launch the developer sets up production hosting on a paid tier (Render paid, or an equivalent managed host) with 99.5% monthly uptime in season and maintenance outside 5 am–9 pm Pacific. The hosting account is created in ReNu-Biome's name with ReNu-Biome's payment method; the developer is added as a collaborator and does the configuration. Monthly cost is reported to the owner before it is committed.

A10.1a Domain, set up by the developer. The developer registers or configures the app's domain (app.renu-biome.com as a subdomain of the existing renu-biome.com, or a new domain if the owner prefers), points DNS at the hosting, and installs an auto-renewing TLS

certificate. If renu-biome.com's DNS is managed by the website host (Shopify), the owner grants the developer access to add records there; the domain itself stays registered to ReNu-Biome.

A10.1b Email sending (SMTP), set up by the developer. Phase 1 sends all customer updates by email; text messaging moves to Phase 2 because US carriers require A2P 10DLC business registration before texting from a business number, which takes days to weeks. The developer sets up a transactional email account (Postmark, SendGrid, Amazon SES or equivalent) in ReNu-Biome's name with the company's payment method, a sending address such as notifications@renu-biome.com, and the SPF, DKIM and DMARC DNS records so mail is not marked as spam. Replies go to the owner's support address. The developer verifies delivery to Gmail, Outlook and iCloud addresses before launch. Customers' email addresses are recorded when the account is created. The developer can start the 10DLC registration during Phase 1 so it is approved in time for Phase 2.

A10.2 Database and backups. Confirm the backend uses a managed database, not in-memory mock data (several routes return fixed records). Nightly backups with point-in-time recovery; one restore test before launch.

A10.3 Monitoring. Error tracking on web and API, an uptime check that alerts the owner, and a status page.

A10.4 Environments. Separate staging and production with their own data; demo accounts only on staging.

A10.5 Ownership and handover. ReNu-Biome owns the app after launch and the developer is on a fixed price, so handover is part of the price: the repository, hosting, domain and messaging accounts in ReNu-Biome's name with the developer as collaborator; infrastructure as code; a runbook for deploys, restores and rotating keys; a README good enough for another developer to take over.

A10.6 Checklist of accounts and assets ReNu-Biome provides for Phase 1. The developer cannot finish without these; the owner or CEO supplies or authorises each. The Phase 2 list is at the end of Part B.

- [ ] Invoice PDFs: the office exports each invoice from QuickBooks as a PDF to upload; no QuickBooks connection is needed at launch.
- [ ] Domain: the owner says which name to use (app.renu-biome.com or other) and grants the developer DNS access; the developer sets it up (A10.1a).

- [ ] Email sending: the owner approves the sending address (for example notifications@renu-biome.com) and the provider's cost, and grants DNS access; the developer creates the email account in ReNu-Biome's name and configures SMTP with the SPF, DKIM and DMARC records (A10.1b).
- [ ] The support inbox support@renu-biome.com, read by the office (decided; no phone number).
- [ ] A weather API account: skipped for now; the weather chip on Home stays hidden until one exists (A4.3).
- [ ] Hosting and database: the owner provides a company payment method; the developer creates the accounts in ReNu-Biome's name, adds themselves as collaborator and configures them (A10.1).
- [ ] Source code repository owned by ReNu-Biome (GitHub organisation or similar), developer as collaborator.
- [ ] Logo files in light and dark versions, the app icon, the QR code and the stock photo: supplied in renu-biome-brand-assets.zip with this document. Brand colours teal #0C6054, orange #F07E42.
- [ ] Product images: the developer takes them from the product pages on renu-biome.com; ReNu-Biome supplies better ones later if wanted.
- [ ] The product list with descriptions and analysis (no prices): the blend list in Part B section 9.1, supplied; OMRI status still to come from the owner.
- [ ] A list of current customers with contact name, email, mobile number, ranches and blocks, so accounts can be created without the developer typing them. Not yet available; the office enters the pilot customer first (Part B section 7.2) and the rest later.

Items moved from the first Phase 1 draft to Phase 2 (Home, accounts, communication, import)

These sections were in the first draft of the February scope and move to Phase 2 on the owner's instruction. Exceptions already pulled back into Phase 1: the sign-in page and Home are built to prototype v2 now (Part B section 8), and the demo logins are disabled on production (Part B section 1.7).

A4. Home screen (UX-5)

A4.1 Remove every hardcoded value. "Sierra Orchards, all entities", the Green Nitrogen and KTS contract bars, "Season 2026, almond", "Post-harvest window is open on N-4 and S-1" and "One recommendation is waiting" are static. Nothing on Home is a literal string except labels.

A4.2 What Home shows. As prototype v2 and Part B section 8.3: greeting with the customer's first name and the ranch weather chip; next delivery (date, program, gallons, tank) or "No deliveries scheduled"; balance due with the count of open invoices; season strip with crop and current stage from a stage table per crop (almond: bloom, soil blends, potassium, hull split, post-harvest); planted ranch cards; quick actions Call ReNu-Biome, Deliveries, Invoices, Ranches. No contract card and no order buttons.

A4.3 Weather at the ranch. Deferred: no weather account for now. Hide the weather chip on Home (do not show "Weather unavailable"); keep the ranch location fields so it can be added later with a provider in the company's name, showing the ranch with the next delivery.

A4.4 Dashboard API. /api/app/grower/dashboard returns zeros for every metric. Make it compute from the customer's records and return nextDelivery, openInvoiceCount, balanceDue, mainCrop, currentStage and ranchWeatherLocation.

A5. Accounts and security (ACC-4, NFR-1, NFR-3)

Login, roles and token expiry work. The scope is simpler than the build assumes: one user per customer, created by ReNu-Biome.

A5.1 Remove the demo logins. The Admin, Emp and Cust one-tap buttons on the sign-in page, and the mock accounts behind them, must not exist in any build a customer can reach. Keep them behind an environment flag for staging only.

A5.2 ReNu-Biome creates every account. Remove the Sign Up link and the Apple and Google buttons. Admin creates a customer, enters the one user's name, email and mobile number, and sends an invitation email with a set-password link that expires in 7 days. Password reset by email.

A5.3 Two roles. Admin (ReNu-Biome owner and office person, two logins) and Customer. Remove the Employee role and its task list from the launch build (flag, section A9).

A5.4 Two-factor on new devices. Email code at login from a device not seen before; no other 2FA at launch since there are no payments in the app. The MFA email code is already built.

A5.5 View as customer. Admin can open any customer's portal read-only with a banner showing whose view it is. No audit log of this is required at launch.

A5.6 Cross-tenant isolation test. Create two real customer accounts; with each one's token, call every /api/app route and confirm no record of the other customer is returned, including by guessing IDs. Make this an automated test that runs on every deploy.

A5.7 Security headers and hardening. Add HSTS, Content-Security-Policy, X-Frame-Options and X-Content-Type-Options on both the API and the web app; rate-limit /api/auth/login; shorten the access token from 7 days to hours with a refresh token. Do this with Part B section 1.

A5.8 Data statement. One paragraph at first login: the customer owns their ranch and block data and ReNu-Biome uses it only to serve them. Export of blocks and deliveries as CSV under More.

A7. Communication (COM-2, COM-4)

Customers email the office. The app's job is to make that one tap away and to deliver status updates the way each customer wants.

A7.1 Email button. An Email ReNu-Biome button on Home, on a delivery and under More, opening the customer's mail app to support@renu-biome.com; Help under More shows that address and what to put in the email (orders: product, gallons, ranch and tank; tank or delivery problems: ranch and tank plus a photo; invoice questions: the invoice number). Decided: no phone number and no hours anywhere in the app.

A7.2 Notification channels per customer. Under More → Notifications, the customer turns on text, push and email independently. Push requires the app installed to the home screen (Part B section 6). Email is the Phase 1 channel.

A7.3 Inbox. Keep the inbox for system events (delivery scheduled, out for delivery, delivered, invoice posted, payment recorded, year-end summary ready; with Part B, sample received, recommendation sent, tank below reorder level). Hide any other category behind its flag.

A7.4 No announcements and no expected response time. Do not build a campaign tool; do not show "we reply within N hours".

A7.5 In-app message thread is behind a flag (section A9). The compose button on Messages is hidden at launch.

A11. Spreadsheet import of blocks (FLD-1)

In Phase 1 the office enters ranches and blocks one at a time from the admin console. Phase 2 adds Import on Ranches: a CSV or XLSX with one row per block (the fields in A1.1) creates ranches and blocks and reports rows it could not read. Export produces the same layout. Import and Export buttons exist today; confirm what they do and point them at this format.

Part A acceptance tests

Run on a phone, with the flags off.

- [ ] Admin creates a customer and its one user from the admin console and gives the customer the password by phone; the customer logs in on a sign-in page that matches prototype v2.
- [ ] Admin enters two ranches and six blocks by hand with crop, variety, acreage and APN; the customer sees them under Ranches with acreage computed per ranch, and no map screen exists.
- [ ] Admin creates an order from a phone call: product, gallons, tank, blocks, window, gate code; the customer receives an email and sees it under Deliveries as Scheduled.
- [ ] Admin marks it Out for delivery then Delivered with the delivered gallons; the customer gets an email at each step; the delivery shows the delivered quantity.
- [ ] The office enters the invoice number, dates, amount and uploads the QuickBooks PDF; it appears in the customer's Invoices linked to the delivery with product and gallons, and the PDF opens.
- [ ] The office marks the invoice Paid with a date; the customer sees Paid and the balance updates.
- [ ] The customer downloads a year-end purchase summary PDF listing the delivery and invoice.
- [ ] The admin Today view lists tomorrow's delivery with gate code and contact as bullets, and the overdue total.
- [ ] Shop, order request, reorder, cancel, recommendations, documents, messages compose, employee tasks and the map are not visible to the customer; admin can see them listed under Feature flags, all off.
- [ ] With the phone in airplane mode, Ranches, a block, Deliveries and Invoices still display.
- [ ] No stock photos, no "fghdhsrhh", no console errors on load.

- [ ] For the items moved to Phase 2: the invitation email arrives for a new user and a login from a second device asks for an email code; Home for a new customer shows their name, ranch weather, "No deliveries scheduled" and a zero balance; the customer turns on email and off text and the next update arrives by email only; View as customer shows exactly the customer's screens; a spreadsheet of blocks imports with errors reported; a second customer's token cannot read the first customer's data; the sign-in page has no demo buttons, Sign Up link or Apple and Google buttons.

Part B. Phase 2

Everything below is the second round of work, built on a working Phase 1. Section numbers 1 to 9 are unchanged from the earlier Phase 2 document.

1. Transfer hosting and ownership to ReNu-Biome

Goal: every account the app depends on is owned by ReNu-Biome, the developer works as a collaborator, and ReNu-Biome could change developers without losing anything. Order matters: the repository first, then infrastructure, then DNS, with a cut-over at the end.

1.1 Source code. ReNu-Biome creates a GitHub organisation (owner: rgundavelli@gmail.com) with two-factor authentication required. The developer transfers both repositories (frontend and backend) into it using GitHub's transfer, which keeps history, branches and issues, rather than pushing a copy. The developer is added as a collaborator with write access, not admin. Branch protection on main: pull request required, no force push. Done: both repos show the organisation as owner and the developer can still push a branch.

1.2 Render. ReNu-Biome creates a Render team on a paid plan with its own payment method and adds the developer as a member. The developer moves both services (web app and backend) into the team, upgrades them from free to a paid instance so there are no cold starts, and connects them to the new organisation repositories so deploys trigger from main. Done: both services run in the ReNu-Biome team, respond within two seconds from cold, and deploy from the organisation repos.

1.3 Database. Confirm what the backend stores data in today. If it is in-memory or a file on the free instance, data is lost on every deploy; this transfer is the moment to move to a managed Postgres (Render Postgres or equivalent) in the team, with daily backups turned

on. The developer migrates the current customers, ranches, blocks, tanks, deliveries and invoices into it. Done: a deploy no longer resets data; a backup from the previous day can be restored to a test instance.

1.4 Secrets and environment variables. Every API key, JWT secret, database URL and email credential is re-entered as a Render environment variable in the team, and the old values are rotated so nothing the developer's personal account held still works. The developer gives ReNu-Biome a list of every secret, what it is for, and where it is set. Done: the list exists and the old JWT secret is invalid.

1.5 Domain and DNS. The app runs at app.renu-biome.com with the API at api.renu-biome.com (decided). The owner grants the developer access to DNS where renu-biome.com is managed; the developer adds the records, sets custom domains on both Render services and confirms the auto-renewing certificates. The onrender.com addresses keep working as aliases. Done: the app loads at the ReNu-Biome domain over HTTPS.

1.6 Email sending. The transactional email account (Postmark, SendGrid or equivalent) is created in ReNu-Biome's name with the SPF, DKIM and DMARC records on renu-biome.com, and the backend sends from support@renu-biome.com (decided); replies land in the same support inbox, which the office reads. If an account already exists under the developer, it is transferred or recreated. Done: a delivery status email arrives in Gmail, Outlook and iCloud inboxes, not spam.

1.7 Environments. Two Render environments: staging (demo logins allowed, test data) and production (demo logins disabled by environment variable, real data). The staging-only demo buttons visible today must not render in production. Done: production sign-in shows no demo buttons.

1.8 Monitoring and backups. Error tracking (Sentry free tier or equivalent) on both services in the ReNu-Biome account; a Render health check and an uptime monitor that emails the owner; database backups verified once. Done: the owner receives a test alert.

1.9 Handover package. A README in each repo with local setup, deploy steps, where each secret lives, how to restore a backup, how to add a customer, and the list of third-party accounts with who owns them. Done: a developer who has never seen the project can run it locally from the README.

1.10 Cut-over. Done in one agreed window outside delivery hours: announce to the office, deploy to the new production, repoint DNS, verify the acceptance test below, keep the old services for seven days, then delete them.

What ReNu-Biome does: creates the GitHub organisation and Render team, provides a payment method, grants DNS access, approves the email sending address, and names an owner for each account. What the developer does: everything else above.

2. OptimoRoute for delivery notifications

Goal: the office plans routes in OptimoRoute as it does today, and the app's delivery statuses and the customer's notifications follow what the driver actually does, without anyone retyping anything. Turns on the existing optimoroute feature flag.

What OptimoRoute's API offers (from its public documentation). Every request carries an API key generated in the OptimoRoute web app under Administration → Settings → WS API; HTTPS only, 5 concurrent requests per account. Orders are created or updated with `create_order` (one at a time, with address geocoding) or `create_or_update_orders` (up to 500, coordinates required); key fields are `orderNo`, `date`, `type` (D for delivery), `location`, `duration`, `timeWindows`, `priority`, `load1` – `load4`, `customFields`, `email`, `phone` and `notificationPreference` (dont_notify, email, sms or both). `get_routes` returns each driver's route for a date with stops, `scheduledAt` and `arrivalTimeDt`; `get_scheduling_information` returns planned times and a `liveEstimate` of arrival from the routes most recently sent to drivers; `send_routes` dispatches a day's routes and can trigger OptimoRoute's own customer notifications. The documentation also lists Get Order Completion, Update Order Completion, Get Mobile Events and Search Orders, whose field detail did not load in my check; the developer must confirm them against the live documentation before estimating. No webhooks are documented, so the app must poll.

2.1 Push deliveries to OptimoRoute. When admin creates or edits a delivery in the app, the backend creates or updates the matching OptimoRoute order: `orderNo` = the app's delivery ID (DEL-101), `date` = the delivery date, `type` D, `location` = the ranch's delivery address or the tank's coordinates (add an address and optional lat/long to the ranch and tank records), `timeWindows` from the delivery window, `duration` by product and gallons, `load1` = gallons, and `customFields` for product, tank, gate code and contact on arrival so the driver sees them on the stop. Cancelling a delivery in the app deletes the OptimoRoute order. Done: a delivery created in the app appears in OptimoRoute's planning view with the right date, window and notes.

2.2 Pull status back. A scheduled job every 5 minutes on delivery days calls the completion and scheduling methods for open deliveries and maps OptimoRoute's state onto the app's statuses: assigned to a sent route → Scheduled with the planned arrival

window; driver started the route or en route to this stop → Out for delivery with the live estimated arrival; completed successfully → Delivered with the completion time, driver name and any proof-of-delivery note, signature or photo the driver captured; failed or rejected → stays Out for delivery and raises an admin alert with the reason. The app never overwrites a status admin set by hand to Cancelled. Done: marking a stop complete in the driver app changes the customer's delivery to Delivered within 5 minutes.

2.3 Customer notifications. Decided: both channels. The app stays the sender of customer emails (the Phase 1 status emails) and adds the live arrival estimate and the OptimoRoute completion time to them; and orders the app pushes carry notificationPreference = sms with the customer's mobile number, so the customer also gets OptimoRoute's own SMS tracking messages (route started, arriving, completed). Build the choice as an admin setting (email, SMS, both) so it can be changed without a release; OptimoRoute's SMS costs are on ReNu-Biome's OptimoRoute plan.

2.4 Delivered quantity. On completion, if the driver recorded a delivered quantity in an OptimoRoute form field, use it as the delivered gallons; otherwise the office enters it as in Phase 1. Proof-of-delivery photos and signature are stored on the delivery and shown to the customer.

2.5 Admin view. The Deliveries tab shows the OptimoRoute driver, planned and estimated arrival, and a sync indicator (last sync time, errors). Today's Operations shows live estimated arrivals for today's stops. A delivery that exists in OptimoRoute but not in the app is listed for admin to link or ignore.

2.6 Settings and failure handling. API key stored as a Render environment variable in ReNu-Biome's account. If OptimoRoute is unreachable, the app keeps working with the last known status and shows a warning in admin; nothing blocks creating a delivery. Dependencies: ranch delivery address (new field), tank coordinates (optional, new field), the Phase 1 delivery statuses.

3. Otodata tank fill monitoring

Goal: the tank level the app shows is the real level from the Otodata monitor on the customer's tank, the office sees which tanks are getting low before the customer calls, and the customer sees the same number. Turns on the existing telemetry feature flag; the tank record already has level and pressure fields holding mock values.

What is known about Otodata's data access. Otodata monitors report by cellular (CAT-M and NB-IoT) to Otodata's Nee-Vo portal and app, and Otodata provides a Web API with a Device model that includes a tank status field (Otodata update note). The API documentation is sent by Otodata to API users on request (support@otodata.ca, 1-844-763-3344); it is not published. Third parties that integrate, such as FuelCloud, have the customer generate an API key in the Otodata portal under Integrations → API Key from an account with admin access, and they note readings can arrive 30 minutes or more after the monitor records them and that 1–2% level drift from temperature is normal. Otodata's site does not list liquid fertilizer among its tank contents, so ReNu-Biome should confirm with Otodata that the monitor model used on its customers' fertigation tanks reports through the same portal and API.

3.1 Get the API documentation first. Before estimating, ReNu-Biome requests the Nee-Vo Web API documentation from Otodata and the developer confirms three things: that readings per device can be fetched by API key (level as a percentage and, if available, volume in gallons, with the reading timestamp), how far back history is available, and whether Otodata can push readings or alerts (webhooks) or must be polled.

3.2 Link monitors to tanks. In the admin console, each tank gets an optional Otodata device serial (or device ID) and the account's API key is stored once as an environment variable. One ReNu-Biome Otodata account covers all monitored tanks; if some customers own their own Otodata accounts, the tank record stores that customer's key instead. Done: admin pastes a serial on a tank and the next reading appears.

3.3 Pull readings. A scheduled job every 30 minutes (or Otodata's webhook, if offered) fetches the latest reading per linked device and stores level percent, gallons (level × tank capacity when Otodata gives percent only), and the reading time. Keep a history of readings per tank. If a device has not reported for 24 hours, mark the tank "no recent reading" rather than showing a stale number as current.

3.4 Show it. Customer: on the ranch screen, each tank shows level percent, estimated gallons, and "as of" time; the Home screen shows the lowest tank for that customer. Admin: a Tanks view across all customers sorted by level, with the reading time, and the same on the customer screen. Replace the mock level and pressure values; drop pressure unless the monitor reports it.

3.5 Low-level alerts. Per tank, a reorder threshold in percent set by admin (default 25%, decided). When a reading crosses below it, email the office and the customer once (decided), and list the tank under Today's Operations as "needs a delivery" with a Create Order shortcut pre-filled with the customer, ranch, tank and gallons to fill. No repeat alert until the level has risen above the threshold again.

3.6 Usage trend. From the reading history, show gallons used per week and an estimated days-to-empty on each tank, so the office can plan deliveries before the alert fires. Keep this simple: average of the last 14 days.

3.7 Failure handling. If Otodata is unreachable, show the last reading with its time; never block any other function. Log sync errors to the admin console. Dependencies: Otodata API documentation and key (ReNu-Biome provides), tank records from Phase 1.

4. Programs per ranch

Goal: ReNu-Biome's nutrition programs (for example pre-harvest, post-harvest, bloom, nut fill) are a thing in the app, so that every delivery belongs to a program, the customer can see what the program calls for and what has been delivered against it, and the office can see which ranches are behind.

4.1 What a program is. A program is a named plan for one crop stage on one ranch for one season. Fields: name (pre-harvest, post-harvest, bloom, nut fill, hull split, dormant, or custom), crop, season year, start and end dates (or the stage window), the blocks it covers (default: all blocks of that crop on the ranch), status (Planned, Active, Complete), notes for the customer. A ranch can have several programs in a season; they can overlap. ReNu-Biome defines a template per crop and stage (products and rate per acre) and the app copies it onto a ranch, where the office can adjust it.

4.2 Program lines. Each program has one or more lines: product, rate per acre, number of applications, timing note ("within two weeks of harvest"). Planned gallons for the program = rate × covered acreage × applications; the office can override. No prices anywhere.

4.3 Deliveries belong to programs. When admin creates a delivery, they pick the program it serves (optional, defaulting to the active program for that ranch and crop). Delivered gallons accumulate against the program line.

4.4 Customer view. On the ranch screen, a Programs section lists this season's programs with status, dates, the blocks covered, the planned products and rates, and a progress bar of delivered versus planned gallons per product. Tapping a program shows its deliveries. The Home screen shows the active program for each ranch and whether it is on track (delivered within the window) or behind (window closing with less than the planned amount delivered). Past seasons' programs stay visible for history.

4.5 Admin view. Programs tab: all ranches with their current programs, filterable by stage and status, with a "behind" indicator. Create a program from a template, edit lines, mark complete. Templates are edited under Products or Settings.

4.6 Reminders. Two weeks before a program's window opens, email the office a list of ranches whose program is still Planned, so deliveries get scheduled. Optionally email the customer when their program becomes Active, with what it calls for.

4.7 Fit with the sample feature. A custom recommendation from a soil or tissue sample (section 5) is saved as a program (or as changes to the active program's lines), so the recommendation and the deliveries that fulfil it are tracked in one place. Dependencies: blocks with crop and acreage (Phase 1, in place); products catalogue.

5. Soil and tissue sample analysis for custom recommendations

Goal: a customer can send ReNu-Biome their lab results (soil or tissue) through the app, ReNu-Biome reviews them and replies with a custom nutrition recommendation the customer can read in the app, and the whole exchange is kept on the block. Turns on the labResults and recommendations feature flags; /api/app/lab-results already exists and returns an empty list.

5.1 Customer submits a sample result. Under a ranch (or from a block), "Send lab results": the customer picks the block(s), the sample type (soil, leaf tissue, petiole, irrigation water), the sample date, the lab name, uploads the lab's PDF or photo (several files allowed), and adds a note ("yellowing on the north end"). Submitting creates a Sample record with status Received and emails the office. The customer sees it listed on the block with its status.

5.2 Office enters the values. In the admin console, the sample opens with the file alongside a form for the key values so the app can chart them: for tissue, N, P, K, Ca, Mg, S, Zn, B, Mn, Fe, Cu (percent or ppm as the lab reports); for soil, pH, EC, organic matter, nitrate-N, P, K, Ca, Mg, Na, CEC; for water, nitrate-N, EC, pH, bicarbonate. Values are optional; the office can save the sample with only the file. Status becomes Under review.

5.3 Reference ranges and flags. A reference table per crop and sample type (almond July leaf, pistachio July leaf, grape bloom and veraison petiole) with low and high bounds, editable by admin under Settings. Start with the values in prototype v2 (almond July leaf: N 2.2 to 2.6%, P 0.10 to 0.30%, K 1.6 to 2.4%, Ca 2.0 to 4.0%, Mg 0.25 to 0.80%, Zn 18 to 40 ppm, B 30 to 65 ppm, Mn 30 to 80 ppm; orchard soil: pH 6.0 to 7.5, EC 0 to 2.0 dS/m, organic matter 1.5 to 4%, nitrate-N 10 to 30 ppm, P 10 to 30 ppm, K 150 to 250 ppm, Ca 1,500 to 3,500, Mg 100 to 400 ppm, Na 0 to 100 ppm, CEC 10 to 25 meq); ReNu-Biome corrects them from the UC guidelines later. Entered values show against the range with a flag: low, in range, high. The customer sees the same chart once the recommendation is sent.

5.4 ReNu-Biome writes the recommendation. From the sample, the office writes a recommendation: a summary paragraph, then lines of product, rate per acre, applications, timing, and which blocks, with the option to save it as a program on the ranch (section 4) or to update the active program's lines. Sending it sets the sample status to Recommended, emails the customer, and shows the recommendation on the block and under a Recommendations list in More. The customer can reply with a question (an email to the office with the sample reference) and the office can revise the recommendation; versions are kept.

5.5 History and trends. Each block keeps every sample and recommendation by date. For tissue, a trend chart per nutrient across seasons. Admin can see all samples across customers with status and age, and samples waiting more than three business days are highlighted on Today's Operations.

5.6 Later, optional. Reading the lab PDF automatically to pre-fill the values, and drafting the recommendation text for the office to edit, can be added once this manual flow is in use; they are not part of this estimate. Dependencies: blocks (in place), programs (section 4), email sending, file storage for uploads (confirm where files are stored and that they are backed up, as part of section 1).

6. Install by QR code

Goal: a customer scans a ReNu-Biome QR code on an invoice, delivery ticket or business card and, in two taps, has the app on their phone's home screen. No app store is involved; this is the web app installed as a home-screen app, which the manifest and service worker already support.

6.1 The link. The QR code points to https://app.renu-biome.com/install (the domain from section 1). Until that domain is live, a temporary code pointing to the current onrender.com address can be used for internal testing only; nothing printed for customers should carry the onrender.com address.

6.2 The install page. A single page at /install, no sign-in required, with the ReNu-Biome logo and three things: the tagline "Farmer's Pride" with "GROWER PORTAL" beneath it, as on the sign-in page, device-specific install steps, and a Sign in button. The page detects the phone: on iPhone (Safari) it shows "Tap Share, then Add to Home Screen" with the share icon drawn; on Android (Chrome) it triggers the browser's install prompt with an "Install app" button and falls back to "Menu → Add to Home screen"; on a desktop it shows

the address to open on a phone and the same QR code. If the app is already installed, the page skips straight to sign-in. Done: a first-time iPhone and Android user each reach a home-screen icon within two taps of scanning.

6.3 The icon and name. The home-screen icon is supplied in the asset pack (renu-biome-app-icon-1024/512/180/64.png): a teal tile with a leaf and the logo's orange tick. Use the 512 in the manifest (purpose any maskable), the 180 as the iOS apple-touch-icon and the 64 as the favicon. The app name under the icon is "ReNu-Biome"; "Farmer's Pride" is the tagline, not the name. Opening from the icon launches full screen without browser chrome; the manifest's display mode is standalone and theme_color is #0C6054.

6.4 The QR code itself. The approved design is below: teal-green modules in the logo's colour (#0C6054), orange (#F07E42) in the three corner markers, the full-colour ReNu-Biome logo on a white badge in the centre, high error correction so the badge does not stop it scanning, and the caption "Scan to install the ReNu-Biome app" with the address. It encodes the plain URL https://app.renu-biome.com/install, not a redirect service, so it never expires. Supplied at 300 dpi; the developer regenerates it from the same design if the URL changes and uses the same two colours for the install page and the app icon.



6.5 Where it goes. The QR code and the sentence "Scan to install the ReNu-Biome app" on the invoice PDF footer and the year-end summary (both generated by the app), and in the delivery status emails as a small image. Printed materials are ReNu-Biome's.

6.6 Later, optional. A listing in the App Store and Google Play (native builds from the existing Expo project) would let a QR code point to the stores instead. It needs an Apple developer account ($99 a year) and a Google Play account ($25), privacy and support

pages, store listings, App Review, and ongoing build work with each release; roughly two to three weeks the first time. Not part of this estimate.

7. From demo to installable: CEO and first customer

Goal: the app stops being a demo. The CEO and one real customer each install it on their own phone from the QR code, sign in with their own account, and see their own data. This is the gate for everything else in this document and the first real test of Phase 1.

7.1 Production environment with no demo content. Production (section 1.7) runs with the demo logins disabled, the mock customers, ranches, deliveries, invoices and the "SRavanga" test product removed, and the real product catalogue loaded. Staging keeps the demo data for testing. Done: a fresh production database contains only ReNu-Biome's real records.

7.2 Two real accounts. The office creates, from the admin console, (a) the CEO as an admin user and (b) the pilot customer with their single login. The pilot customer is the almond grower modelled in the owner's program workbook (section 9): about 1,400 acres, N target 165 and K_2O target 135 units per acre. Their ranches, blocks and tanks are entered by hand from the customer list ReNu-Biome provides, and their 2026 season is loaded as delivered history from the workbook's Applications tab: the four soil-blend applications (March 5, March 17–23, April 7–8, May 9), Fruit Care (May 25–26), the potassium acetate applications (April 3–4, June 13, June 15, July 8–10) and the bloom foliars, each with its product and gallons, so the customer's first view shows a full season rather than empty screens. The 2027 almond program from the workbook is created on their ranch as the first program once section 4 exists. Passwords are given by phone. Done: both can sign in on production with no help from the developer, and the pilot customer's Deliveries tab shows the 2026 applications with the right products and gallons.

7.3 Install from the QR code. The install page (section 6.2) is live at app.renu-biome.com/install and the QR code from section 6.4 scans to it. The CEO installs on an iPhone and the customer on whatever phone they use; the icon, name and full-screen launch work on both. Done: both phones show the ReNu-Biome icon and open the app from it.

7.4 One real cycle. The office creates a delivery for the first customer, marks it out for delivery and delivered, enters the invoice with its QuickBooks PDF, and later marks it paid; the customer receives the email at each status change and sees the delivery, invoice and PDF in the app. Done: the Phase 1 acceptance script passes on production with real data.

7.5 Support path. Decided: the customer emails support@renu-biome.com if something fails; the office reads that inbox and emails the developer, with a response time for the pilot period agreed before the customer is onboarded. The developer watches the error tracker daily for the first two weeks. No phone line and no published hours.

7.6 Order of work. This section comes before the OptimoRoute, Otodata, programs and sample features: those are built against a working production app, not a demo. Suggested sequence: section 1 (hosting), then section 6 and this section (install and pilot), then sections 2 to 5.

8. Login and Home page, from the prototype

Goal: the sign-in and Home screens, and the look of every other screen, match the clickable prototype ReNu-Biome approved: grower-portal prototype v2 (October 21, 2026) at https://renu-grower-portal-preview.netlify.app/v2/, also supplied as grower-portal-prototype-v2.html. v2 already applies the Phase 1 scope rules and shows the Phase 2 features marked "Phase 2"; the switch under More → Prototype v2 hides them to show the February release alone. The September design (v1, at https://renu-grower-portal-preview.netlify.app/v1/) stays online for comparison only; where v1 and v2 differ, v2 wins. Sign in to v2 with the prototype shortcuts or any email with the password demo.

8.1 Colours and type. Use the logo's teal-green #0C6054 where the prototype uses dark green, the logo's orange #F07E42 as the single accent (current stage dot, highlighted items), white cards with hairline borders and large rounded corners, soft (tinted, not solid) buttons as in the prototype, and the Manrope typeface or the system font. Replace the prototype's forest green throughout the app so the app, the install page and the QR code share one green.

8.2 Sign-in page. Full-screen photo of an almond orchard in bloom (the prototype's stock image is used for now, in the asset pack; ReNu-Biome replaces it with a licensed or own photo before production) under a teal gradient darkest at the top and bottom; the ReNu-Biome logo centred near the top; a white card with Email, Password (with show/hide), a Sign in button, and a "Forgot password?" link that emails a reset link; the sentence "Need an account? Email support@renu-biome.com" instead of any sign-up link (decided: no phone number anywhere in the app). No demo buttons on production (section 7.1). Errors read in plain words ("That password doesn't match"). Done: the page matches the prototype's look on a phone and a laptop, and a wrong password shows the message without clearing the email.

8.3 Home page. The prototype's photo header: the bloom photo edge to edge from the top of the screen, teal shade so text stays readable, the logo top-left, a time-of-day greeting with the user's first name, the customer's name beneath, and a frosted chip with the current crop stage and the orange dot (the weather chip is deferred, A4.3; the slot stays for it). The season card overlaps the bottom edge of the header: crop and year, a five-segment stage strip with the current stage highlighted, and one line of status ("Post-harvest window is open on N-4 and S-1" comes from the active program when section 4 exists; before that, the stage name only). Then "This week": next delivery (date, product, gallons, tank), any recommendation or sample waiting (when those flags are on), and balance due with the count of open invoices. Then "Planted ranches" cards (ranch, county, acres, blocks, tanks, and the lowest tank level once Otodata is live). Quick actions: Email ReNu-Biome (a mailto link to support@renu-biome.com), Deliveries, Invoices, Ranches. Bottom navigation: Home, Ranches, Deliveries, Invoices, More, in the prototype's pill style. Nothing on Home is a hardcoded string except labels; empty states say what is empty ("No deliveries scheduled").

8.4 Left out from the prototype, on purpose. The entity switcher, the 2026 price agreement card, Request an order and Reorder, the message thread, the global search bar, the Spanish toggle and the Company admin entry. The More screen keeps Documents, Notifications, Your data and Help only as far as those exist in the phase being built.

8.5 Loading and offline. As in the prototype: skeleton placeholders while data loads, the last-loaded Home cached for offline reading, and the screen transitions the prototype shows (slide in forward, slide back) where the framework allows them cheaply.

8.6 What v2 adds to the reference (October 21). Branding: the tagline "Farmer's Pride" centred under the logo on the sign-in and install pages, white with a short orange rule beneath it and "GROWER PORTAL" as a small bold tracked label; the logo at twice the v1 size on sign-in; a frosted (translucent, blurred) sign-in card over the bloom photo, the photo slightly more saturated with a warm orange glow in its lower half. Lists: every delivery row shows gallons and the program name ("1,976 gal Post-harvest"), never the blend analysis; the analysis appears only on the delivery detail. Admin Today: each metric tile carries an icon (truck, calendar, invoice, tank, flask, programs), and each delivery card leads with gallons and program name, then the date, blocks, gate code, contact, driver and ETA as bullet points. Design language throughout: section labels as small uppercase tracked text, status tags with a leading dot, a gradient primary button, a frosted bottom navigation bar, soft shadows and hover states on cards and tiles. The install page uses the light-background logo (teal wordmark, orange tick), never a tinted copy of the white logo.

8.7 Admin console in v2. The prototype's admin console, with Owner and Office roles and the tabs Today, Deliveries, Customers, Programs and More (Invoices, Products, Samples, Tanks, Program templates, Feature flags, Reports, Settings), is the reference for the Phase 1 admin items in Part A (section A6) and for the Phase 2 admin views in sections 2 to 5. Today includes Mark delivered → delivered gallons → invoice entry with the QuickBooks PDF, and Create order from a phone call with the suggested gallons from the active program. The v1 CEO and CTO dashboards are withdrawn.

Done: side by side on a phone, the developer's sign-in and Home are recognisably the prototype's screens with the items in 8.4 removed and real data in every field.

9. What the owner's program spreadsheet means for the app

The owner's workbook (ReNu_vs_Traditional_2026) models the pilot customer's season: 1,400 acres, targets of 165 units N and 135 units K_2O per acre, the 2026 applications with dates and gallons, a 2027 program by stage, an ILRP nitrogen-balance check, and a cost comparison against UAN32 and CAN17 programs. Four of its tabs are the data the app's programs and reporting need; the pricing tabs stay outside the app. This section says what to take from it and where it lands.

9.1 The real product catalogue. The Inputs tab lists what ReNu-Biome sells; the build's catalogue (Biome Care, N-CARE, K-RUSH, Bee Bloom) is placeholder and is replaced by this list. Each product needs an analysis the app can calculate with: percent N, percent K_2O, percent Ca where present, and density in lb per gallon, so that gallons delivered convert to pounds of nutrient. Decided: this blend list is the catalogue; the website's line names (Green Nitrogen, Peptide Nutrition, Bee Bloom and so on) are kept as a brand label on the matching blend, not as separate products. OMRI status per product is not yet known; the OMRI flag stays off until the owner confirms.

Product	lb/gal	% N	% K_2O	Notes
ReNu N-32%, K-28% Blend	13.98	32	28	Soil blends, 1st to 4th application; 2027 blends carry Ca at 15%, 10%, 20%
ReNu N-32%, K-24% Blend (K acetate)	13.98	32	24	3rd application

Product	lb/gal	% N	% K_2O	Notes
ReNu N-32%, K-28% Blend + B, Zn	14.12	32	28	4th application, with boron and zinc
ReNu Fruit Care	13.97	32	28	Additional blend shot (2026 only)
ReNu 0-0-29 potassium acetate	12.25	0	29	Soil or foliar
ReNu N-6%, K-16%, Ca-20% Blend	11	6	16	Post-harvest
Bloom foliars + boron	—	0	0	Dormant, bee bloom, bloom, petal fall; no N or K credit

No prices are entered in the app (owner's decision stands); the $/ton and $/gal columns stay in the owner's workbook. Add to the product record: percent Ca, percent B and Zn where applicable, and a "counts toward N" flag (bloom foliars do not).

9.2 Program templates come straight from the ReNu 2027 tab. It is exactly the program structure in section 4: ordered applications, each with a product and a rate in gallons per acre. The almond template is: bloom foliars + boron (3 gal/acre, season); soil blend 1st app (8 gal/acre, early spring); soil blend 2nd app (8); soil blend 3rd app with K acetate (8); soil blend 4th app (8); potassium acetate soil/foliar (3.5); custom K acetate blend (1.3); foliar K acetate (0.5); post-harvest blend (2.6). Planned gallons for a ranch = rate × block acreage, as section 4.2 says. The owner's stage names (bloom, 1st to 4th soil application, post-harvest) replace the generic ones in section 4.1. Pistachio and grape templates: not yet available from the CEO. Create them as placeholders (one program per stage with the stage names Bloom, Nut fill, Hull split, Post-harvest for pistachio and Bud break, Bloom, Veraison, Post-harvest for grape, no lines) so the office can fill in products and rates from the admin console without a release.

9.3 Nutrient accounting per acre. The Summary and Applications tabs compute, per application and for the season, pounds of N and K_2O delivered (gallons × lb/gal × percent), delivered units per acre, and percent of target. The app can produce this from its own records once products carry analysis: for each ranch and season, N and K_2O applied per acre from deliveries, against the crop target (165 N and 135 K_2O for almonds; pistachio and grape targets are not yet known, so the template fields are left blank and the progress bar shows gallons only until the office enters them; targets are a field on the program

template). This is the progress bar in section 4.4, expressed in units per acre as well as gallons. Add an "effective N" line using the loss factor the owner uses (6% for ReNu blends), labelled as ReNu-Biome's estimate.

9.4 ILRP nitrogen balance (the compliance flag, defined). The NMP tab is the Irrigated Lands report check: N applied (A) from deliveries, N removed (R) = kernel yield × 68 lb N per 1,000 lb, and the A/R ratio against the Central Valley Water Board's acceptable range for mature almonds (about 1.43 to 1.44). The removal factor and A/R range for pistachio and grape are not yet known; store them per crop in Settings, blank until the office enters them. When the compliance flag is turned on, the app computes A per block and ranch from deliveries (plus any irrigation-water nitrate the customer enters), takes R from the yield the customer enters per block, and shows A, R, A/R and the status against the range, with the per-APN export the coalition needs. Yield per block per season (Phase 1 list, FLD-4) becomes a required field for this.

9.5 Delivery log fields. The Applications tab records, per application, the product, date range, gallons and pounds of nutrient. The app's deliveries already hold product, gallons, date and blocks; adding the computed lb N and lb K_2O to each delivery record (and to the CSV export) lets the office rebuild this tab from the app without a spreadsheet.

9.6 What stays out of the app. The traditional-program comparison, N-loss cost, market blend cost and break-even tabs are a sales tool for the owner and contain prices. They do not go in the customer portal. If the owner later wants a customer-facing "season report" (N and K_2O delivered vs target, gallons hauled, A/R status), it can be built from 9.3 and 9.4 without any dollar figures; note it as a later option.

9.7 Data from the workbook. The workbook itself is not supplied to the developer; everything the app needs from it is in this document: the product list with analyses (9.1), the almond program template with rates and stage windows (9.2), the almond targets and N-loss factor (9.3), the almond N-removal factor and acceptable A/R range (9.4), and the pilot customer's 2026 application dates and products (7.2). Gallons per application for the pilot are computed as rate × block acreage once the office has entered the blocks.

9.8 Data model notes for a later Salesforce import. ReNu-Biome's Salesforce org holds 17 customer accounts, 12 owner contacts, 25 farms (ranch level only, crop on 16, acreage on 2), 45 soil and tissue analyses with the lab PDFs attached, 89 delivery trips (gallons and driver, no dates, free-text location) and a five-item brand catalogue. It is not imported in Phase 2: the pilot customer is entered by hand per 7.2, and the Salesforce export is redone when the office decides to bring the other customers across. So that the import does not fight the schema when it comes, the data model carries four things now:

- Customer and Ranch each keep an optional `externalId` (the Salesforce Account Id and Farm Id). An import matches on it and updates instead of creating duplicates; the office never sees it.
- A lab result has a `sampleType` (soil, tissue or water) and an attached PDF, and the typed values (pH, EC, N, P, K, result and reference) are optional. The PDF is the record of truth; 17 of the 46 Salesforce analyses are tissue reports stored in the soil object with zeros in the soil fields. This is the same model section 5 uses for customer-submitted samples.
- The products catalogue is the blend list in 9.1 with analysis fields (% N, % P_2O_5, % K_2O, % Ca, lb/gal), not the five brand names in Salesforce (Biome-Care, N-Care, Bee Bloom, K-Rush, Sea Biolife). If the owner wants the brand names kept, they are a `brand` label on the blend, not separate products.
- A delivery can be saved as history without an order or program: date, ranch, gallons, driver, optional product lines, flagged `historical`. That lets the trip log land later once the office adds dates and picks the ranch each free-text location means, and it is the same shape 7.2 uses for the pilot's 2026 applications.

Nothing else from Salesforce needs a home in the app: opportunities are Salesforce sample data, the recommendation records are PCA PDFs that attach to a ranch as documents, and the state fertilizer licences are company compliance records that belong on the admin page if anywhere.

What ReNu-Biome provides

The developer cannot start several items without these. Each names who supplies it.

- [ ] Done: GitHub organisation created with 2FA, owner rgundavelli@gmail.com, developer invited as collaborator.
- [ ] Render team on a paid plan with the company payment method: created by the owner; invitation sent to the developer (sravangaaismartsolutions@gmail.com) on October 21. The developer accepts it and moves both services in (section 1.2).
- [ ] DNS for renu-biome.com: app.renu-biome.com and api.renu-biome.com exist only once records are added where the domain's DNS is managed (Shopify's Domains settings, or the registrar). Either the owner grants the developer access to that one area, or the developer sends the owner the records to paste in: a CNAME for app and one for api pointing at the Render services, plus Render's certificate verification record (owner, section 1.5).

- [ ] Decided: notifications send from support@renu-biome.com; the owner approves the email provider's cost when the developer reports it (owner).
- [ ] OptimoRoute API key from Administration → Settings → WS API in ReNu-Biome's OptimoRoute account, and confirmation of which plan features are on (Proof of Delivery, Order Tracking notifications). Waiting on the CEO; section 2 cannot start without it (owner).
- [ ] A ranch delivery address and, where known, tank GPS coordinates for every customer, so OptimoRoute can route to them. Not yet available; the office enters them per ranch from the admin console, pilot customer first (office).
- [ ] Otodata Nee-Vo Web API documentation and an API key from the Otodata portal (Integrations → API Key, admin account), plus the device serial for each monitored tank and confirmation from Otodata that the fertigation-tank monitors report through the same API. Waiting on the CEO; section 3 cannot start without it (owner; contact support@otodata.ca).
- [ ] Program templates: almond supplied (section 9.2); pistachio and grape still to come from the CEO, built as empty placeholders until then (owner).
- [ ] Reference ranges for tissue and soil values: start with the prototype's values (section 5.3); corrections from the UC guidelines later (owner).
- [ ] The list of labs customers use and a sample report from each, so the entry form matches what growers will upload. Not yet available; build the form to the prototype's fields and adjust later (office).
- [ ] Decided: customer delivery notifications by both the app's email and OptimoRoute's SMS tracking messages (section 2.3).
- [ ] Decided: low-tank threshold default 25%; alerts go to the office and the customer (section 3.5).
- [ ] Completed Phase 1 items that this phase depends on: invoice PDF upload, the real product catalogue, and the Phase 1 acceptance test run (developer and office).
- [ ] Supplied: app icon artwork at 1024, 512, 180 and 64 px and the approved QR code, in renu-biome-brand-assets.zip (section 6).
- [ ] The approved prototype v2 at https://renu-grower-portal-preview.netlify.app/v2/ (and the grower-portal-prototype-v2.html file), with v1 at /v1/ for comparison (owner, section 8).
- [ ] From the program workbook: the product list with analyses, the almond program template with rates and stage windows, per-crop N and K_2O targets, the N-loss factor, the N-removal factor and A/R range per crop, and the pilot customer's 2026 applications (owner, section 9).

- [ ] Nothing from Salesforce for Phase 2. When the office is ready to bring the other customers across, a fresh Salesforce export with trip dates filled in and each trip location matched to a ranch (office, later).

Acceptance tests

Run on a phone against production (hosting) or staging (features), with real ReNu-Biome accounts and no demo logins.

Hosting transfer

- [ ] Both repositories are owned by the ReNu-Biome GitHub organisation; a push to main by the developer deploys to Render inside the ReNu-Biome team.
- [ ] The app loads at the ReNu-Biome domain over HTTPS, and within two seconds after an hour idle.
- [ ] Production sign-in shows no demo buttons; staging does.
- [ ] A deploy does not reset data; yesterday's database backup restores to a test instance.
- [ ] The old JWT secret no longer authenticates; the secrets list matches the Render environment.
- [ ] A status email from notifications@renu-biome.com lands in Gmail, Outlook and iCloud inboxes.
- [ ] The owner receives a test alert from the uptime monitor.
- [ ] A second developer, given only the README, runs the backend and frontend locally.

Install and pilot

- [ ] Production holds no demo logins, mock customers or the test product; staging still does.
- [ ] The CEO scans the QR code on an iPhone, installs from the install page in two taps, and signs in with their own admin account.
- [ ] The first customer scans the same code, installs on their phone, signs in with the login the office created, and sees only their own ranches, blocks, tanks, deliveries and invoices.
- [ ] A real delivery for that customer runs Scheduled → Out for delivery → Delivered with an email at each step, and its invoice with PDF appears and is later marked Paid.
- [ ] The office knows whom to call, and the developer confirms the error tracker shows no errors from the pilot phones.

Login and Home

- [ ] The sign-in page shows the orchard photo, logo, email and password, Forgot password and the ReNu-Biome phone number; no sign-up link, no demo buttons on production; a wrong password shows a plain message and keeps the email.
- [ ] Home shows the photo header with greeting, customer name, weather at the ranch and the current stage; the season strip; next delivery; balance due; ranch cards; and the four quick actions, all from real data, with none of the items in 8.4 present.
- [ ] Every green in the app, the install page and the QR code is #0C6054 and the accent is #F07E42.

OptimoRoute

- [ ] A delivery created in the app appears as an order in OptimoRoute with date, window, gate code and contact on the stop.
- [ ] After the route is sent to the driver, the app shows Scheduled with the planned arrival; when the driver starts, Out for delivery with a live estimate; when the driver completes the stop, Delivered within five minutes with the completion time and driver.
- [ ] The customer receives one email per status change and no duplicate from OptimoRoute (or the reverse, per the owner's choice).
- [ ] Cancelling in the app removes the OptimoRoute order; a failed stop raises an admin alert and does not mark Delivered.

Otodata

- [ ] A tank linked to a device serial shows the level and reading time that the Otodata portal shows, within one reading cycle.
- [ ] Dropping a tank below its threshold (or lowering the threshold below the current level) produces one email to the office and the customer and a "needs a delivery" entry on Today's Operations with Create Order pre-filled.
- [ ] A device silent for 24 hours shows "no recent reading"; the app otherwise works with Otodata unreachable.

Programs

- [ ] Admin creates a post-harvest almond program on a ranch from the template; the customer sees it on the ranch with blocks, products, rates and zero progress.
- [ ] A delivery assigned to that program moves its progress bar; the Home screen shows the program as on track or behind according to the window.
- [ ] Two weeks before a Planned program's window, the office receives the reminder email.

Samples and recommendations

- [ ] A customer uploads a tissue PDF for block N-4 with a note; the office receives an email and sees it as Received.
- [ ] The office enters values; potassium below the almond July leaf range shows a low flag.
- [ ] The office sends a recommendation saved as a program change; the customer gets an email, sees the recommendation and the chart on N-4, and the program lines reflect it.
- [ ] A sample older than three business days without a recommendation is highlighted on Today's Operations.